

Label-Free Compositional DGA Representations and a Surveillance-Bias Caution for Transformer Risk Screening

Thibaut Faucheux

CESI Engineering School, France

KMUTNB, Bangkok, Thailand

Email: thibautfaucheux1@gmail.com

Dr. Rattanakorn Phadungthin

College of Industrial Technology

KMUTNB, Bangkok, Thailand

Abstract—Dissolved Gas Analysis (DGA) interpretation for power transformers remains rule-based (IEC ratios, Rogers, the Duval Triangle), expert-dependent, and label-scarce. This paper makes three contributions, learned *without any fault labels*, on a longitudinal field dataset of 4563 samples from 628 transformers (2019–2024). First, we show that representing each sample by the centred log-ratio (CLR) of its combustible-gas composition separates fault *type* from gassing *severity* by construction: clustering a two-dimensional *linear* projection of the CLR recovers Duval fault-type structure (adjusted Rand index 0.55 vs 0.16 for raw concentrations), while a neural autoencoder (SD-CAE) and an adversarial disentanglement term add nothing over this linear pipeline—negative results we report. Second, and importantly, we show that the field-event notes commonly used to validate such models are *confounded by surveillance bias*: a trivial sampling-count predicts logged events as well as the gas chemistry does, and a label-free risk score loses its apparent forward validity once this is controlled—a caution for the DGA machine-learning literature, for which we give a confound-free chemistry-based evaluation. Third, we build a label-free health/risk screening index (severity, acetylene-aware, temporal, and anomaly terms) that outranks conventional severity baselines, reported together with the confound floor. Our results indicate that label-free compositional representations are a useful basis for DGA fault-type discovery, and that—at least in this fleet—a trivial sample count can dominate operator-logged event labels, so DGA risk models validated on such labels should report a sample-count confound floor.

Index Terms—label-free representation learning, dissolved gas analysis, compositional data analysis, fault diagnosis, label bias, transformer condition monitoring

I. INTRODUCTION

Power transformers are among the most critical and expensive assets in an electrical network, and an in-service failure can cause extended outages and high repair costs. Dissolved Gas Analysis (DGA) is the most widely used technique for the early detection of incipient faults: thermal and electrical stresses decompose the insulating oil and paper, releasing characteristic gases—hydrogen, methane, ethane, ethylene, acetylene, and the carbon oxides—whose relative amounts indicate the type of fault [1], [2].

Conventional interpretation methods, namely the IEC 60599 ratios, the Rogers ratios, and the Duval Triangle, are rule-based. They encode expert knowledge as fixed thresholds,

require an expert to apply and arbitrate, and return a “not determined” result for a non-negligible share of samples [2], [3]. Supervised machine learning can reach very high accuracy; a recent example fuses the conventional methods into a single score and feeds an ensemble classifier, reporting 99.17% accuracy [4]. Such methods, however, require labelled fault data—and often the conventional rules themselves as inputs—which utilities rarely have at fleet scale.

This paper studies a label-free alternative on a real operational dataset of 4,563 main-tank samples from 628 transformers (2019–2024), and makes three contributions. We first observe that representations built on gas concentrations encode overall gassing *magnitude* rather than fault *type*, and show that the centred log-ratio (CLR) of the combustible-gas composition separates the two by construction, raising agreement with the Duval fault types from an adjusted Rand index of 0.16 to 0.55 with a simple linear projection; a neural autoencoder (SD-CAE) and an adversarial disentanglement term add nothing, which we report as negative ablations. We then show, second, that the field-event notes routinely used to validate such models are, in this fleet, *confounded by surveillance bias*: a trivial sample count out-predicts the gas chemistry, and a label-free score built from each unit’s early history loses its apparent forward validity once this count is controlled—a caution we support with a confound-free, chemistry-defined evaluation. Third, we build a label-free health/risk screening index that outranks conventional severity baselines, reported together with this confound floor.

The contributions of this work are:

- a label-free *severity-disentangled compositional representation*: the centred log-ratio of the combustible-gas composition, on which a linear two-dimensional projection recovers Duval fault-type structure (ARI 0.16 $\rightarrow$ 0.55); the neural (SD-CAE) and adversarial-independence variants are reported as negative ablations;
- a finding that, in this fleet, the field-event label used to validate such models is confounded by surveillance bias—a trivial sample count out-predicts the gas chemistry and the apparent forward validity vanishes once it is controlled—together with a confound-free, chemistry-

defined evaluation target;

- a label-free health/risk screening index (severity, acetylene-aware, temporal, and anomaly terms) that outranks conventional severity baselines, reported against the confound floor.

The remainder of the paper is organised as follows. Section II reviews related work; Section III describes the dataset and preprocessing; Section IV the method; and Sections V–VI the experiments and results, before the conclusion.

II. RELATED WORK

We organise prior work along the trajectory mapped by a recent 34-page survey [5]: rule-based interpretation, supervised machine learning, and unsupervised / representation learning. We then add the point that motivates this paper: how such models are *validated*.

Conventional DGA interpretation. The dominant methods are rule-based: the key-gas method, the IEC 60599 and Rogers gas ratios, and the Duval Triangle and Pentagon [1]–[3], [6], [7]. They are simple and interpretable but rely on fixed thresholds, require expert judgement, and leave borderline cases “not determined”. The current IEEE guide bases its gas-level norms on the 90th and 95th percentiles of a large population and embeds the Duval Triangle as an interpretation aid [1].

Supervised machine learning for DGA. Most learning-based work casts fault diagnosis as supervised classification, using neural networks [8], support vector machines [9], [10], and more recently ensembles and deep models; the survey reports accuracies rising from $\sim 85\text{--}95\%$ (classical ML, few hundred samples) to $>97\%$ (deep learning, ≥ 5000 samples) at the cost of opacity and data hunger [5]. A representative recent system fuses the four conventional methods into a single “AI-score” feature and reaches 99.17% [4], but it requires fault labels *and* the conventional rules as inputs. The recent transformer *Health Index* literature is likewise supervised: it predicts an expert-defined health score, so it needs that score as a label. The survey names prognostics and remaining-useful-life estimation as the open frontier.

Unsupervised and representation learning. Autoencoders [11] and variational autoencoders [12] learn compact representations without labels, and statistical-ML reviews of DGA exist [13]. Closest to our work, Wang et al. [14] apply a sparse autoencoder to DGA, but a supervised back-propagation head performs the final classification and the input is the IEC ratios on 134 samples. Liu et al. [15] cluster DGA with a correlation-coefficient DBSCAN, but tune amplification coefficients using known fault categories on 60 class-balanced samples. A common thread is that fault *type* is read from per-sample gas *proportions* (Duval percentages, IEC ratios, Liu’s relative contents), which discards magnitude—an observation we make explicit and exploit. Dukarm et al. make the geometric setting explicit by treating the gas proportions as points on a simplex, generalising the Duval triangle to four dimensions [16], but retain Euclidean coordinates; to our knowledge the log-ratio

TABLE I
PER-GAS SUMMARY OVER THE 4,563 SAMPLES (CONCENTRATIONS IN PPM).

Gas	Median	Max	% zero
H2	16	2,537	1.1
CH4	18	2,469	10.1
C2H2	0	954	97.4
C2H4	2	3,394	39.1
C2H6	14	1,085	9.5
CO	193	2,437	0.8
CO2	2,281	23,574	0.7

(Aitchison) geometry of compositional data analysis [17] has not previously been applied to DGA.

Gap and the validation problem. Two gaps motivate this work. First, existing “unsupervised” DGA still depends on labels (for a classifier head or for parameter tuning) and on small, curated, balanced datasets; we study an end-to-end label-free pipeline on a real, imbalanced operational fleet. Second, and more subtly, label-free risk and health models are routinely validated against operator-logged field events. We show that, in our fleet, this validation is *confounded by surveillance bias*—units that operators watch closely are both sampled more and more likely to have a logged event—so a trivial sampling count predicts logged events as well as the gas chemistry does. This phenomenon is well documented in epidemiology and health-records machine learning as surveillance or informed-presence bias [18]–[20]; to our knowledge it has not been reported for DGA. We quantify it, adapt the standard control—conditioning on the encounter count [20]—to transformer risk validation, and accordingly lead with a confound-free fault-type result and provide a confound-free evaluation target.

III. DATASET AND PREPROCESSING

Dataset. We use DGA records from transformer main tanks: 4,563 samples from 628 transformers and 42 manufacturers, sampled between 2019 and 2024 (median six samples per unit over ~ 4.5 years). Rated voltages are recorded as free-text multi-winding strings (e.g. “115/22”) and are not used as model inputs. A per-gas summary is given in Table I.

Data provenance. The dataset was assembled for this study from operational main-tank DGA records of in-service transformers in Thailand (multiple operational sources). Unit identifiers were anonymised, and no nameplate or location information is used. The raw data are not publicly available; an anonymised extract (hashed unit identifiers, dates, gas concentrations, and binary event flags) is released with our code for reproducibility.

Features. Each sample provides seven diagnostic gases used as model features: H2, CH4, C2H2, C2H4, C2H6, CO and CO2. Atmospheric oxygen and nitrogen, and the total-combustible-gas sum, are excluded to avoid redundancy and leakage.

Data characteristics. After cleaning, the gases have essentially no missing values; sparsity instead appears as zeros—

acetylene is zero in 97.4% of samples, consistent with arcing being rare. The concentrations are heavily right-skewed and remain skewed for C2H2 and C2H4 even after a logarithmic transform. One oxygen reading is a clear data-entry error (about 7.2×10^8 ppm), which motivates outlier clipping.

Preprocessing. The pipeline is configuration-driven: each gas is clipped at its 99.9th percentile, missing readings are imputed as zero (taken as below the detection limit), a $\log(1+x)$ transform is applied, and the features are standardised.

Conventional labels (weak, external). For comparison we compute Duval Triangle 1, IEC 60599 and Rogers labels with our reference implementation. The Duval class distribution (over all samples) is imbalanced: PD 1511, T1 1451, T2 686, T3 492, D2 30, D1 24, DT 16, with about 7.7% not determined. Many of the PD samples are methane-dominant cases that Triangle 1, lacking a stray-gassing zone, labels PD (the Duval Pentagon adds such a zone); this is mostly a per-sample effect – at the per-unit (latest-state) level only about 100 units are PD. We therefore treat the conventional labels as indicative, not ground truth.

Validation notes. 391 of 4,563 samples (8.6%) carry a raw free-text field-event note (for example a Buchholz trip or a bushing failure). Filtered to genuine protection/damage events and aggregated to units, these give the weak unit-level label `fault_note` used for validation (Section V); the sample-level 8.6% and the unit-level label rate are therefore different quantities. Most notes are partly in Thai (69%; 127 are Thai-only) and invisible to an English keyword filter; a curated translation of all 211 distinct Thai notes (released with the code, pending domain validation) identifies 15 additional event notes, whose effect is evaluated in Section VI.

IV. METHODS

We present three label-free components: a compositional representation that recovers fault *type* (Section IV-A); a health/risk screening index (Section IV-B); and an evaluation protocol that exposes and controls a surveillance-bias confound in the field labels (Section IV-C). Each sample provides seven diagnostic-gas concentrations (H2, CH4, C2H2, C2H4, C2H6, CO, CO2); the fault-type composition below is built from the five combustible gases, while the carbon oxides feed only the severity term (Section IV-B).

A. Severity-disentangled compositional representation

A plain autoencoder on gas concentrations encodes overall gassing *magnitude* rather than fault *type* (Section VI). We separate *how much* gas is present from *which* gases dominate. Writing $\mathbf{x} \in \mathbb{R}_{\geq 0}^5$ for the five combustible gases (H2, CH4, C2H2, C2H4, C2H6), the total gassing and a magnitude term are

$$g = \sum_{i=1}^5 x_i, \quad m = \log(1 + g), \quad (1)$$

and the composition $\mathbf{p} = \mathbf{x}/g$ lies on the simplex $\mathcal{S}^4$. Since Euclidean operations are not meaningful on the simplex, we

map $\mathbf{p}$ to Euclidean space with the centred log-ratio (CLR) transform of compositional data analysis [17],

$$\mathbf{c} = \text{clr}(\mathbf{p}), \quad c_i = \log \frac{p_i}{\left(\prod_{j=1}^5 p_j\right)^{1/5}}, \quad \sum_i c_i = 0. \quad (2)$$

CLR is undefined at zero, and the data contain many true zeros (C2H2 is zero in 97.4% of samples), so zeros are handled by a multiplicative replacement with constant δ before the log-ratio; δ is treated as a sensitivity-checked hyperparameter (Section VI). A fault-type code is then any map $\mathbf{z} = f(\mathbf{c})$ of the standardised CLR coordinates: because f sees only the composition, $f(\text{clr}(\lambda \mathbf{x})) = f(\text{clr}(\mathbf{x}))$ for any $\lambda > 0$ —the same fault type at any gassing level maps to the same code, severity invariance *by construction*. We deploy the simplest instantiation: a two-dimensional principal-component projection of the standardised CLR. As ablations we also instantiate f as a small autoencoder (*SD-CAE*), minimising the CLR-space reconstruction loss $\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_n \|g_\phi(f_\theta(\mathbf{c}_n)) - \mathbf{c}_n\|_2^2$, optionally with an adversary h_ψ that tries to recover m from $\mathbf{z}$ through a gradient-reversal layer, enforcing statistical independence $\mathbf{z} \perp m$ with strength λ :

$$\min_{\theta, \phi} \max_{\psi} \mathcal{L}_{\text{rec}} - \lambda \frac{1}{N} \sum_n (h_\psi(\mathbf{z}_n) - m_n)^2. \quad (3)$$

Neither neural component helps: the autoencoder does not improve on the linear projection, and since composition and severity are partly, legitimately correlated (arcing is both acetylene-rich and high-gas), over-enforcing independence erases fault signal; both are reported as negative ablations with the achieved $R^2(m \mid \mathbf{z})$ (Section VI). We cluster $\mathbf{z}$ with KMeans at $k = 7$, fixed to the Duval class count for external comparison (the silhouette-selected k is reported as a robustness check), and measure agreement with the conventional Duval classes by the adjusted Rand index (ARI) and normalised mutual information (NMI); the labels are used for evaluation only, never in fitting. The two-dimensional $\mathbf{z}$ is a learned, label-free diagnostic map.

B. Label-free health/risk index

For fleet screening we summarise each transformer’s history into one risk score from *physical gas features only* (we deliberately exclude sample-count features; see Section IV-C). The score is a weighted sum of standardised components: a *severity* term led by hydrogen (the universal early-fault gas) with CO2 and the total combustible gas; an *acetylene* term (arcing, the most dangerous fault, up-weighted); a *temporal* term, the H2 generation rate by multi-point linear regression in ppm/year following IEEE C57.104; and a self-supervised *anomaly* term from autoencoder reconstruction error and an Isolation Forest. The fleet is ranked by this score.

C. Evaluation and the surveillance-bias control

The dataset has no ground-truth fault labels. A subset of samples carry free-text field-event notes; we derive a

weak binary label `fault_note` by matching genuine protection/damage events (trip, Buchholz, differential, sudden-pressure, bushing, overheating), excluding administrative or maintenance entries. We report agreement with this label by the area under the ROC curve (AUC). Crucially, this label is *not* a clean target: operators sample worrisome units more often, so the sample count is itself predictive of a logged event. We therefore (i) always co-report the AUC of the sample count alone as a confound floor; (ii) perform a strict temporal hold-out—features from each unit’s first half of history predict an event in its second half—and residualise the score on the first-half sample count to test whether any forward signal survives; and (iii) introduce a confound-free, chemistry-defined target: the *onset of arcing*, i.e. whether acetylene appears (> 2 ppm) within two years in a unit that currently has none. This target depends on gas chemistry, not operator attention, and is uncorrelated with the sample count.

V. EXPERIMENTAL SETUP

Data and preprocessing. We use the dataset of Section III (4,563 main-tank samples from 628 transformers, 2019–2024; median 6 samples per unit over ~ 4.5 years). Gas values are clipped at the 99.9th percentile, missing readings imputed as zero (below detection), log-transformed, and standardised; SD-CAE additionally uses the CLR composition of the five combustible gases (H₂, CH₄, C₂H₂, C₂H₄, C₂H₆). The weak label `fault_note` is positive for 71 of 628 units (11.3%).

Baselines. For fault-type recovery we compare the deployed CLR-PCA representation against clustering on standardised log-concentrations (the severity baseline) and on raw proportions, KMeans and a Gaussian mixture directly on the 5-D CLR, and the SD-CAE autoencoder with adversary weights $\lambda \in \{0, 1, 10\}$ (all negative ablations). For risk screening we compare the index against conventional severity proxies—total combustible gas, a C57.104-style count of gases above the fleet 90th percentile, and a Duval-arcing flag—and against the sample-count confound. For the anomaly term we compare autoencoder reconstruction, Isolation Forest, and Deep SVDD, plus a battery of standard detectors.

Metrics. Fault-type recovery is measured by ARI and NMI against the Duval classes (a deterministic rule, hence a confound-free target). Risk ranking is measured by AUC against `fault_note`, precision at the top- k units, and the event rate by risk decile; severity leakage of a representation is measured by $R^2(m \mid \mathbf{z})$. Stochastic results are reported as mean $\pm$ standard deviation over multiple seeds; headline AUCs carry unit-level bootstrap 95% confidence intervals, paired comparisons use a paired bootstrap, and the chemistry target of Section VI uses a unit-blocked cluster bootstrap because its positives concentrate in few units.

Implementation. Models are small multilayer perceptrons trained on CPU (PyTorch); the data is small enough that no GPU is required. Every figure and number is produced by versioned scripts writing to a results directory, with fixed seeds for reproducibility. The main hyperparameters are listed in Table II.

TABLE II
MAIN HYPERPARAMETERS.

Component	Setting
Deployed representation	CLR $\rightarrow$ standardise $\rightarrow$ PCA, 2 components
SD-CAE (ablation)	MLP 16–8, ReLU, latent 2; Adam, lr 10^{-3}
SD-CAE training	batch 128, 200 epochs, early stop (patience 20)
CLR zero-replacement δ	10^{-3} (sensitivity: 10^{-4} – 10^{-2})
Adversary weight λ	ablated at 0, 1, 10 (not deployed)
Index weights	severity 1, acetylene 2, temporal 1, anomaly 1
Clustering	KMeans, $k = 7$ (Duval classes); silhouette k as check

VI. RESULTS AND DISCUSSION

A. Fault-type recovery: severity vs type

Representations built on gas concentrations agree only weakly with the Duval fault types (KMeans ARI 0.16; a plain autoencoder 0.14) because they encode gassing *magnitude*: the total gas is linearly readable from the autoencoder’s code ($R^2(m \mid \mathbf{z}) = 0.63$). Replacing concentrations with the CLR composition removes magnitude by construction and recovers fault *type*: KMeans on the standardised 5-D CLR reaches ARI 0.49, and the deployed two-dimensional PCA projection **ARI** 0.545 ± 0.002 (NMI 0.54), versus 0.12 for raw proportions (Fig. 1a). The improvement is therefore due specifically to the log-ratio geometry—not to normalisation, and *not to the neural network*: the SD-CAE autoencoder reaches only 0.47 ± 0.06 , and the adversarial independence term degrades it further (ARI 0.30–0.31 at $\lambda = 1$ –10), confirming that part of the composition–severity correlation is legitimate fault signal. We therefore deploy the linear variant and report both neural components as negative ablations. The structure generalises across time—fitting on pre-2022 samples and assigning 2022+ samples to the frozen centroids yields ARI 0.45—and is stable for zero-replacement $\delta \leq 10^{-3}$ (ARI 0.46–0.55), though it degrades at $\delta = 10^{-2}$, an unavoidable sensitivity when acetylene is zero in 97% of samples; at unit level (latest sample per unit) agreement is 0.35. The two-dimensional code is a learned, label-free diagnostic map on which Duval classes separate visibly (Fig. 1b). Because Duval is a deterministic rule rather than an attention-driven label, this is our most defensible result; the moderate absolute ARI reflects that Duval itself is a heuristic over an imbalanced fleet, so we report the large *relative* gain.

B. Label-free health/risk screening

The risk index ranks the fleet with AUC 0.74 (unit-bootstrap 95% CI [0.68, 0.80]) against `fault_note`, versus 0.52–0.56 for the conventional severity proxies—though a trivial sample-count predictor reaches a statistically indistinguishable 0.76 on this confounded label (Fig. 3a), so the comparison must be read against that floor (Sec. VI-C). Its top decile carries a $\sim 2.4\times$ event rate and its top ten units a $\sim 4.4\times$ rate; the event rate falls from $\sim 27\%$ in the riskiest decile to $\sim 2\%$ in the safest, with the five riskiest deciles above the fleet base rate (Fig. 2). A component ablation shows the order of importance: severity (0.64) $\rightarrow$ +acetylene (0.71) $\rightarrow$ +temporal

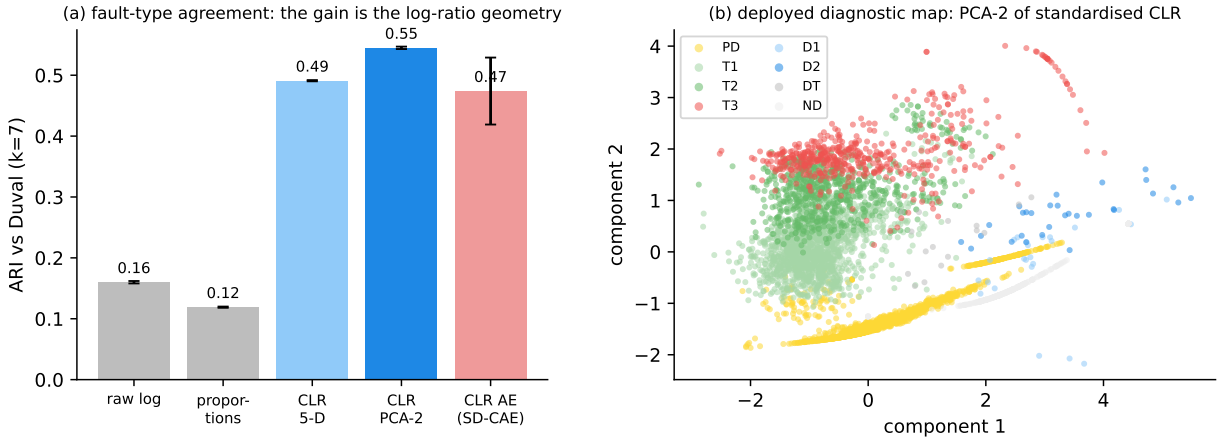


Fig. 1. Severity-disentangled compositional representation. (a) Fault-type agreement with Duval across representations (mean $\pm$ std, five seeds): the gain comes from the CLR log-ratio geometry, and a linear PCA projection beats the SD-CAE autoencoder (negative ablation). (b) The deployed label-free diagnostic map—PCA-2 of the standardised CLR—coloured by Duval class; discharge (D1/D2), partial discharge, and thermal (T1–T3) regions separate.

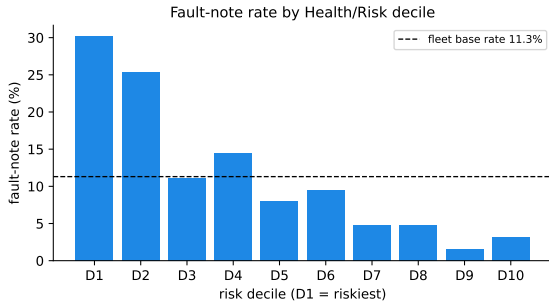


Fig. 2. Field-event rate by health/risk decile (D1 = riskiest). The rate falls from $\sim 27\%$ (riskiest) to $\sim 2\%$ (safest); the five riskiest deciles exceed the fleet base rate (dashed).

H2 rate (0.73) $\rightarrow$ +anomaly (0.74); acetylene-awareness is the dominant lever, consistent with arcing being the most dangerous fault. Among anomaly detectors, Isolation Forest (0.60) and autoencoder reconstruction (0.58) are weak but additive, whereas Deep SVDD collapses on this small tabular feature space (AUC 0.52); a battery of nine standard detectors yields no improvement over the deployed combination. The component set was chosen during development with feedback from this same label, so the absolute AUC carries some design optimism; the component ordering, not the third decimal, is the finding.

C. The field-event label is surveillance-confounded

This is the paper’s cautionary result. A trivial predictor—how many times a unit has been sampled—scores AUC **0.76** against `fault_note`, statistically indistinguishable from the physics index (0.74; paired-bootstrap difference CI $[-0.10, +0.06]$, $p = 0.58$) and far above the conventional baselines (Fig. 3a). The mechanism is operational: utilities re-sample units they are worried about, so the sample count encodes attention rather than gas chemistry, and 69% of the notes contain Thai text largely missed by the English keyword

label. In a joint logistic model the count adds signal far beyond the physics (likelihood-ratio $p = 9 \times 10^{-13}$) while the physics retains a small significant increment ($p = 0.003$): the label is dominated by, though not purely, operator attention. Repairing the label with our Thai-note translation (six additional own-unit electrical events) makes the count’s advantage disappear entirely (AUC 0.734 vs 0.735), indicating that part of its edge came from label incompleteness. The consequence is decisive under a strict temporal hold-out: a label-free score built from each unit’s first-half history predicts second-half events at AUC 0.54, the first-half sample count alone predicts them better (0.68), and once the score is residualised on that count its forward AUC falls to chance (0.50, Fig. 3b). Because the first-half sample count is known *before* the second-half events, this is a prospective effect, not reverse causation. We therefore do not claim genuine forward fault prediction from `fault_note`; any AUC against it must be read against the 0.76 confound floor.

To evaluate without this confound we use a chemistry-defined target—the onset of arcing (acetylene appearing within two years in a unit that has none), which is uncorrelated with the sample count (correlation -0.01 with the target). Across 3,794 prediction points the positives are rare and clustered: 45 positive points arising from only 17 onset units. Accounting for this clustering with a unit-blocked bootstrap, ethylene and hydrogen forecast onset at AUC 0.64 (95% CI $[0.52, 0.77]$) and 0.63 ($[0.53, 0.73]$)—both excluding chance—while the sample count is uninformative (0.49, CI $[0.38, 0.58]$, Fig. 3c); the result is robust across nine onset-threshold and horizon variants (AUC 0.64–0.73). Thermal activity (ethylene) and incipient hydrogen thus precede arcing—a physically sensible, confound-free early signal, modest in effect size and with wide intervals owing to the rarity of the event, but distinguishable from chance.

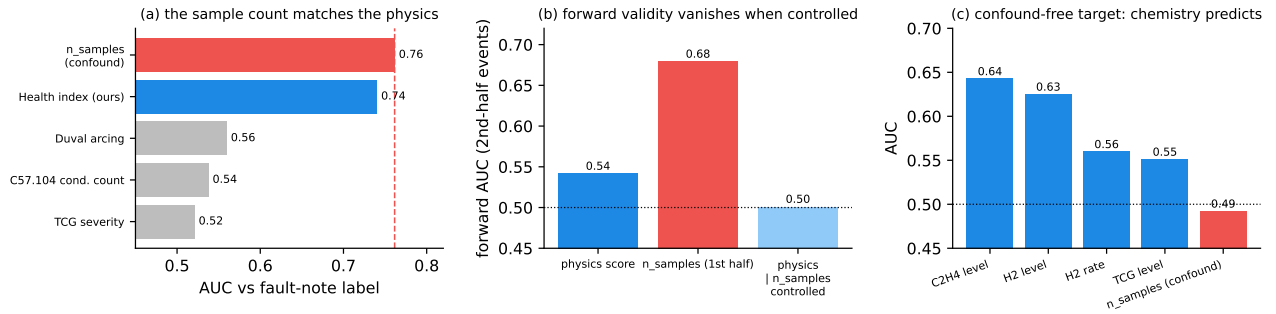


Fig. 3. The surveillance-bias analysis. (a) A trivial sample count matches the physics index against the field-note label and exceeds every conventional severity baseline (the confound floor). (b) Strict temporal hold-out: the score’s forward validity vanishes once the sample-count confound is controlled, while the count alone forward-predicts. (c) Confound-free chemistry target (arcing onset): gas chemistry predicts, the sampling confound does not.

VII. CONCLUSION AND FUTURE WORK

We studied label-free condition monitoring of a real transformer fleet from Dissolved Gas Analysis and reported three results. (1) A severity-disentangled compositional representation—the centred log-ratio of the combustible-gas composition—recovers fault *type* by construction, raising agreement with the Duval classes from ARI 0.16 to 0.55 with a simple linear projection; the neural (SD-CAE) and adversarial-independence variants are documented negatives. (2) The field-event notes commonly used to validate such models are, in this fleet, confounded by surveillance bias: a trivial sample count matches the gas chemistry (AUC 0.76 vs 0.74, difference not significant) and a label-free score built from these features loses its apparent forward validity once that count is controlled. (3) A label-free health/risk index nonetheless outranks conventional severity baselines, and on a confound-free, chemistry-defined target the gas trajectory weakly forecasts the onset of arcing.

Limitations. The dataset has no ground-truth fault labels, so every result is measured against a rule (Duval) or a weak, biased note label; the fault-type agreement is moderate; and the findings come from a single fleet. The methods are deliberately simple—a linear projection in the right geometry and small multilayer perceptrons—rather than a new architecture.

Future work. The highest-leverage improvement is a better target—completing the domain validation of our Thai-note translation (which already recovers 15 missed event notes) or obtaining real failure/maintenance records—after which the risk and prognostic claims can be strengthened. We also see promise in confound-free, chemistry-defined prognostic targets and, following recent surveys [5], in physics-informed extensions. More broadly, we recommend that DGA studies which validate against operator-logged events report the sample-count confound floor, as a trivial surveillance signal can otherwise be mistaken for diagnostic skill.

REFERENCES

- [1] IEEE, “IEEE guide for the interpretation of gases generated in mineral oil-immersed transformers,” IEEE Std C57.104-2019, 2019.
- [2] M. Duval, “A review of faults detectable by gas-in-oil analysis in transformers,” *IEEE Electrical Insulation Magazine*, vol. 18, no. 3, pp. 8–17, 2002.
- [3] International Electrotechnical Commission, “Mineral oil-filled electrical equipment in service — guidance on the interpretation of dissolved and free gases analysis,” IEC 60599:2022, Ed. 4.0, 2022.
- [4] A. G. Saleh, A. E. Azzam, G. Attiya, and E. Ibrahim, “Detecting incipient faults in power transformers through hybrid model of DGA and machine learning,” *Electric Power Systems Research*, vol. 256, p. 112877, 2026.
- [5] G. Ramarao, K. D. Rao, P. Chongala, P. Pavani, S. Dawn, and T. S. Ustun, “Advances in dissolved gas analysis for power transformer diagnostics: From traditional methods to artificial intelligence driven prognostics,” *Engineering Applications of Artificial Intelligence*, vol. 181, p. 115390, 2026.
- [6] M. Duval, “The Duval triangle for load tap changers, non-mineral oils and low temperature faults in transformers,” *IEEE Electrical Insulation Magazine*, vol. 24, no. 6, pp. 22–29, 2008.
- [7] M. Duval and A. dePablo, “Interpretation of gas-in-oil analysis using new IEC publication 60599 and IEC TC 10 databases,” *IEEE Electrical Insulation Magazine*, vol. 17, no. 2, pp. 31–41, 2001.
- [8] Y. Zhang, X. Ding, Y. Liu, and P. J. Griffin, “An artificial neural network approach to transformer fault diagnosis,” *IEEE Transactions on Power Delivery*, vol. 11, no. 4, pp. 1836–1841, 1996.
- [9] K. Bacha, S. Souahlia, and M. Gossa, “Power transformer fault diagnosis based on dissolved gas analysis by support vector machine,” *Electric Power Systems Research*, vol. 83, no. 1, pp. 73–79, 2012.
- [10] S.-w. Fei and X.-b. Zhang, “Fault diagnosis of power transformer based on support vector machine with genetic algorithm,” *Expert Systems with Applications*, vol. 36, no. 8, pp. 11 352–11 357, 2009.
- [11] G. E. Hinton and R. R. Salakhutdinov, “Reducing the dimensionality of data with neural networks,” *Science*, vol. 313, no. 5786, pp. 504–507, 2006.
- [12] D. P. Kingma and M. Welling, “Auto-encoding variational Bayes,” in *International Conference on Learning Representations (ICLR)*, 2014.
- [13] P. Mirowski and Y. LeCun, “Statistical machine learning and dissolved gas analysis: a review,” *IEEE Transactions on Power Delivery*, vol. 27, no. 4, pp. 1791–1799, 2012.
- [14] L. Wang, X. Zhao, J. Pei, and G. Tang, “Transformer fault diagnosis using continuous sparse autoencoder,” *SpringerPlus*, vol. 5, p. 448, 2016.
- [15] Y. Liu, B. Song, L. Wang, J. Gao, and R. Xu, “Power transformer fault diagnosis based on dissolved gas analysis by correlation coefficient-DBSCAN,” *Applied Sciences*, vol. 10, no. 13, p. 4440, 2020.
- [16] J. Dukarm, Z. Draper, and T. Piotrowski, “Diagnostic simplex for dissolved-gas analysis,” *Energies*, vol. 13, no. 23, p. 6459, 2020.
- [17] J. Aitchison, “The statistical analysis of compositional data,” *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 44, no. 2, pp. 139–177, 1982.
- [18] D. L. Sackett, “Bias in analytic research,” *Journal of Chronic Diseases*, vol. 32, no. 1–2, pp. 51–63, 1979.
- [19] E. R. Haut and P. J. Pronovost, “Surveillance bias in outcomes reporting,” *JAMA*, vol. 305, no. 23, pp. 2462–2463, 2011.
- [20] B. A. Goldstein, N. A. Bhavsar, M. Phelan, and M. J. Pencina, “Controlling for informed presence bias due to the number of health encounters in an electronic health record,” *American Journal of Epidemiology*, vol. 184, no. 11, pp. 847–855, 2016.